ALICE L WALTON SCHOOL OF MEDICINE

Hypertension: Foundations and Frameworks

Last updated February 13, 2026 · 14 min read · ScholarRx

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Adapted from: Hypertension: Foundations and Frameworks

Listen to this Brick
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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the different types of hypertension.
2. Describe the clinical presentation of hypertension.
3. Describe the common complications of hypertension.
4. Interpret cli ary hypertension.

CASE CONNECTION

You are seeing CJ, a 56-year-old male, for his second follow-up visit for blood pressure management. Over the past 3 months, despite attention to lifestyle modification, his blood pressure remains elevated with several systolic readings of 140 to 152 mm Hg and diastolic readings of 92 to 98 mm Hg. Home blood pressures are within the same range. CJ has no chest pain, blurred vision, shortness of breath, or headaches. His physical examination, urinalysis, and serum creatinine are normal. You discuss starting antihypertensive medication with CJ. "No way! I don't need to take pills. I feel absolutely fine. You asked me all those questions, and I told you I don't have any of those symptoms. My weight is good, I exercise daily, I watch my diet, and I don't put any salt on my food. Why should I take pills if I feel good? Why fix it if it isn't broken?" he says.

What are some potential underlying pathophysiologic mechanisms of CJ's elevated blood pressure? How will you explain the importance of blood pressure control to him? Consider your answers as you read, and we'll revisit CJ at the end of the brick.

GO TO CONCLUSION ↓

What Is Hypertension? (LO #1)

Hypertension (high blood pressure) occurs when the force of blood exerts too much pressure on the walls of blood vessels. Hypertension increases the risk of serious health proble cranial hemorrhage, aneurysm, and chronic kidney failure, so early diagnosis is crucial. According to the World Health Organization (WHO), 1.13 billion people globally have hypertension; 20% to 25% do not know they have it.

Risk factors for hypertension include age, genetics (family history), using tobacco, obesity, lack of physical activity, stress, and other chronic conditions such as diabetes, chronic kidney disease (CKD), obstructive sleep apnea (OSA), and more.

Diagnosis and Stages of Hypertension

The diagnosis of hypertension can be established with the average of at least two elevated blood pressure readings taken at least 2 weeks apart. Since 2017, **stage 1 hypertension** has been defined as a systolic blood pressure of 130 to 139 mm Hg and/or a diastolic blood pressure 80 to 89 mm Hg. This is the most commonly used definition of hypertension. The other stages are shown in Table 1.

Table 1. Categories of Hypertension

Blood pressure category	Systolic blood pressure, mm Hg	Qualifiers	Diastolic blood pressure, mm Hg
Normal	<120	And	<80
Elevated	120-129	And/or	80-89
Stage 1 hypertension	130-139	And/or	80-89
Stage 2 hypertension	≥140	And/or	≥90
Severe asymptomatic hypertension (formerly known as hypertensive urgency)	organ damage		≥120
Hypertensive emergency	≥180 with accompanying organ damage	And/or	≥120

While higher stages of hypertension portend a worse prognosis in terms of chronic complications, there are also implications for treatment.

Stage 1 Hypertension. Patients may not need medication immediately. A trial of lifestyle changes (eg, reducing weight, sodium intake, and alcohol use) may lower the blood pressure to normal levels.

Stage 2 Hypertension. Patients will likely need to start antihypertensive medications to lower their blood pressure to normal levels, but they should also continue the lifestyle interventions.

Severe Asymptomatic Hypertension. As the blood pressure rises above 180/120 mm Hg, the risk of severe consequences (eg, stroke, myocardial infarction, heart failure) increases. Blood pressure control with oral antihypertensive medication requires frequent outpatient monitoring and dose adjustment as needed.

Hypertensive Emergency. Patients have signs of end-organ damage and require hospitalization to emergently lower the blood pressure with intravenous (IV) antihypertensive treatment.

Types of Hypertension

Hypertension can be primary (idiopathic) or secondary (caused by an underlying disease).

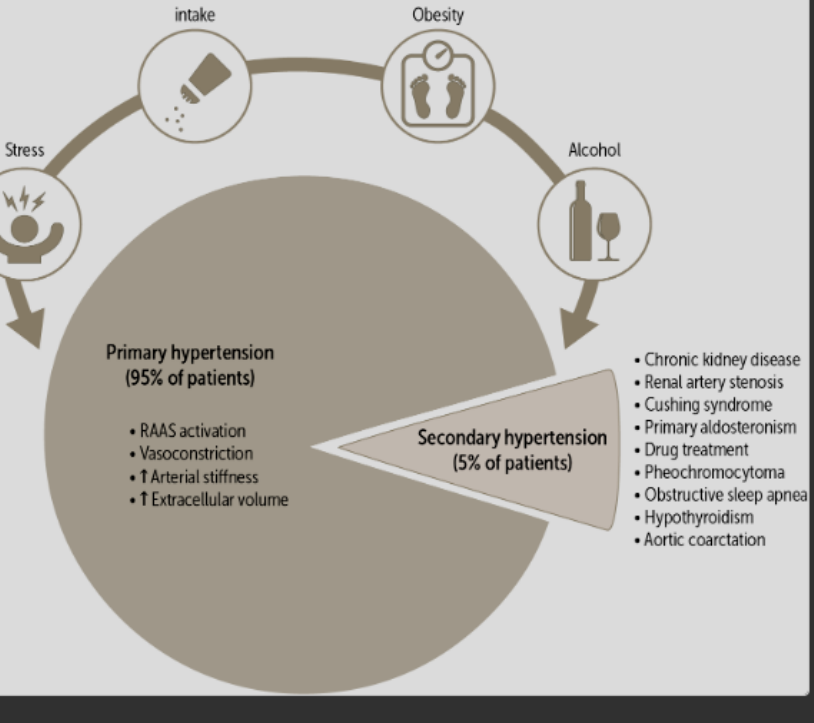


Figure 1. Primary versus Secondary Causes of Hypertension

Regardless of type, environmental factors such as stress, increased sodium intake, obesity, or heavy alcohol intake can further increase the blood pressure.

Primary Hypertension. Also known as essential hypertension, primary hypertension is high blood pressure without an identifiable cause. This type accounts for 95% of cases.

Secondary Hyperter associated with a known medical condition, commonly chronic kidney disease, renal artery stenosis (RAS), or obstructive sleep apnea.

How can we differentiate between primary and secondary hypertension?

Severe Asymptomatic Hypertension. Severely elevated systolic blood pressure (≥180 mm Hg) and/or diastolic blood pressure (≥120 mm Hg) with **no signs of acute organ deterioration or injury** was formerly known as **"hypertensive urgency."** Controlling severe asymptomatic hypertension usually does not require hospitalization, only close management in the outpatient setting.

Hypertensive Emergency. Severely elevated systolic blood pressure (≥180 mm Hg) and/or diastolic blood pressure (≥120 mm Hg) **with signs of acute organ injury** requires immediate medical attention, including hospitalization and intravenous (IV) antihypertensives. Organ damage can include:

- Cardiovascular: heart failure (pulmonary edema), myocardial infarction (MI), unstable angina, and aortic dissection
- Neurologic: stroke, hypertensive retinopathy, and encephalopathy
- Renal: acute kidney injury

Epidemiology

In the US, hypertension is more common in males than females (49.5% men, 43.9% women). Among the middle-aged population, early onset is higher among men than women for Black individuals (56.8% men, 56.7% women) and Hispanic individuals (50.4% men, 36.3% women) when compared to White individuals (47.0% men, 39.0% women). The prevalence of hypertension among Asian men (49.8%) is also higher than among Asian women (39.1%).

Risk Factors

Modifiable risk factors for hypertension include obesity, high sodium and heavy alcohol intake, and low physical activity. Increased salt intake can lead to increased wear retention and hypertension, and about 50% of patients may be salt-sensitive, where this effect is exaggerated. Underlying treatable kidney disease is also a strong risk factor, as most renal diseases are associated with hypertension. Non-modifiable risk factors include increasing age, genetic predisposition (family history), and race (Black patients).

How Do Patients With Hypertension Present? (LO #2)

Patients with hyperte ich may delay the diagnosis and treatment. The high blood pressure is usually picked up incidentally at a routine well visit or at a health fair or employment physical.

Less commonly, when the blood pressure is very high and the patient is experiencing end-organ damage (i.e. hypertensive emergency), symptoms may include:

- Severe chest pain
- Altered mental status
- Blurred vision
- Shortness of breath
- Decrease in the urine output
- Headache

How does the patient with hypertensive emergency present?

What Are Some nsion? (LO #3)

Chronic uncontrolled hypertension can damage the heart, brain, blood vessels, kidney, and eyes.

Heart

Persistent high blood pressure increases cardiac demand, as the heart must pump against high systemic vascular resistance (SVR). The increased afterload increases cardiac work and left ventricular (LV) pressure. To compensate, the left ventricle thickens, leading to left ventricular hypertrophy (Figure 2).

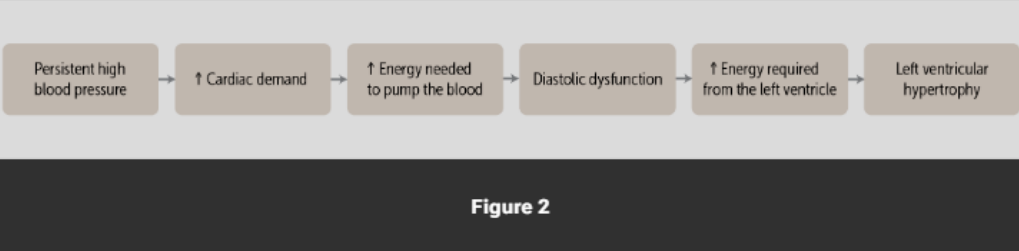
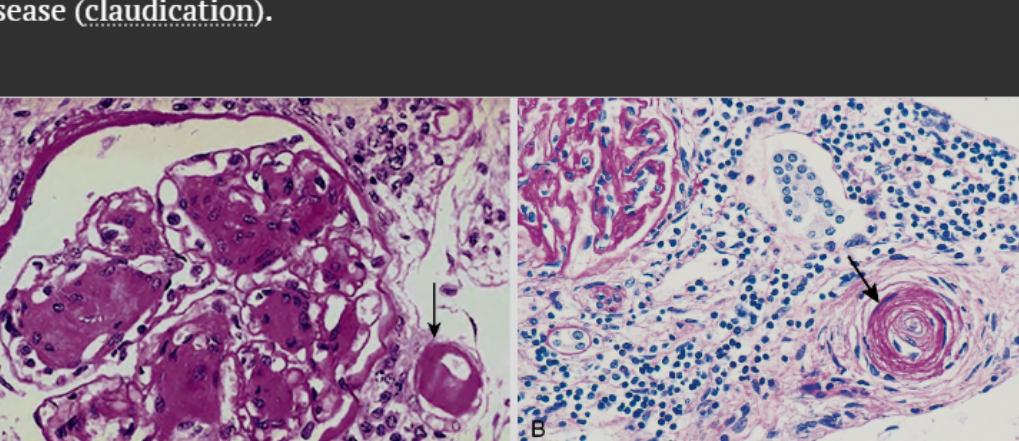

Persistent high blood pressure
↑ Cardiac demand
↑ Energy needed to pump the blood
Diastolic dysfunction
↑ Energy required from the left ventricle
Left ventricular hypertrophy

Figure 2

The thickened LV does not relax well during diastole (**diastolic dysfunction**), and the reduced size of the LV cavity leads to poor filling and reduced stroke volume, causing nonsystolic heart failure.

Blood Vessels

Persistently elevated blood pressure damages the blood vessels, increasing atherosclerosis. The increased pressure causes increased growth of the smooth muscle of blood vessels, increasing these cells' remodeling and hypertrophy (Figure 3). This reduc ood flow and increasing resistance. With time, high blood pressure can lead to vessel microdamage, which later results in aggregation of cholesterol and blood cells as well as the evolution of atherosclerotic plaques, further reducing the gauge of the blood vessel lumen and eventually vessel blockage. This may lead to coronary artery disease (CAD), including MI, stroke, and/or peripheral artery disease (claudication).



Vascular pathology in hypertension. (A) Hyaline arteriolosclerosis. Markedly thickened arteriole (arrow) with protein deposition (hyalinization) and a markedly narrowed lumen is seen to the right of the glomerulus in the kidney. (B) Hyperplastic arteriolosclerosis (onion-skinning) causing luminal obliteration (arrow) (periodic acid–Schiff stain).

How do atherosclerosis and coronary artery disease (CAD) develop?

Brain

Persistent high blood pressure can be the cause of oxidative stress in the form of excessive generation of reactive oxygen species (ROS), decreased capacity of antioxidants, or both. This damages small blood vessels and affects vessel wall integrity. The small arteries can develop lipohyalinosis, a form of small artery wall thickening associated with hypertension, diabetes, and advanced age. Endothelial damage leads to vessel wall thickening and ballooning, often with leakage of serum out of the artery into surrounding tissues. This eventually leads to small vessel occlusion with microinfarction of the surrounding brain (lacunar strokes).

Additionally, uncontrolled hypertension over time can lead to small outpouchings of the vessel wall (microaneurysms) leading to hemorrhagic strokes.

A correlation between cerebral beta amyloid deposits and hypertension has been found, suggesting that dementia may also be associated with hypertension even in the absence of acute strokes.

CLINICAL CORRELATION

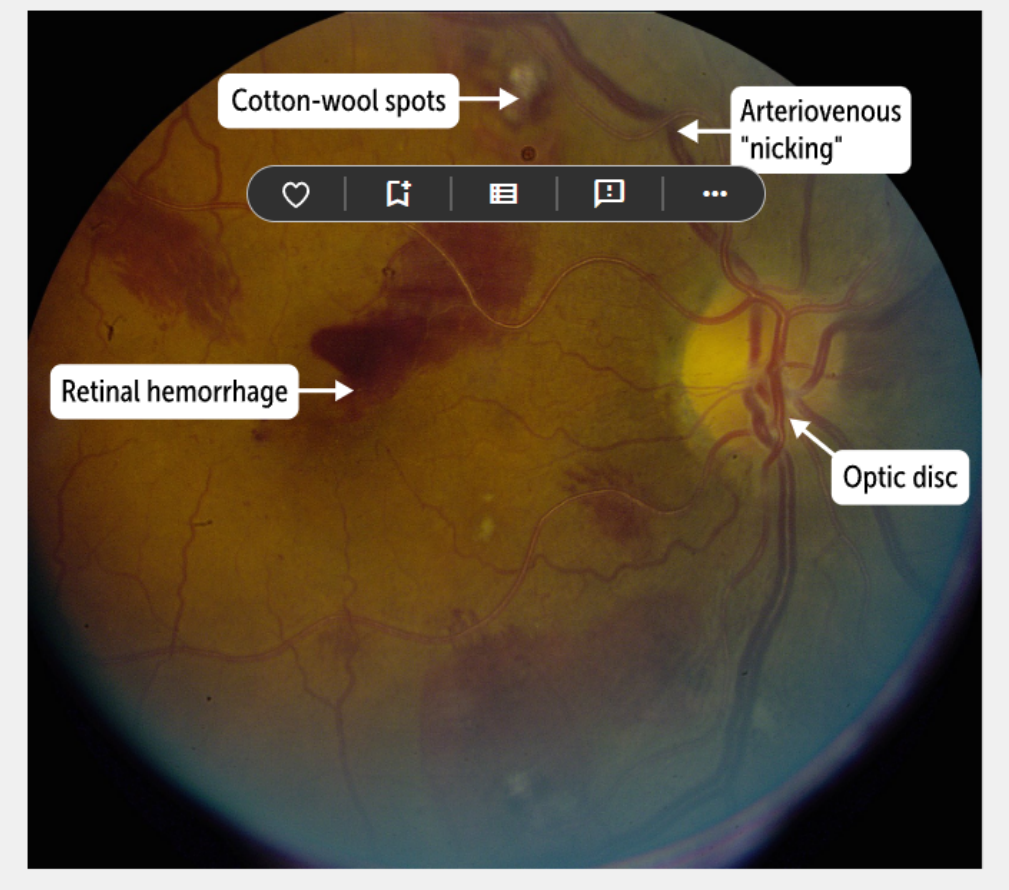
Dementia, particularly Alzheimer's disease, is associated with beta-amyloid deposits in the brain. Beta amyloid is a peptide that is a normal byproduct of brain function; however, in Alzheimer's disease, these peptides are produced in abnormal amounts or are not cleared effectively. These plaques (accumulations of beta-amyloid deposits) disrupt normal brain function by interfering with neuron signaling and can trigger inflammation, ultimately leading to neuron death, brain shrinkage, and cognitive impairment like memory loss.

Kidney

Uncontrolled high blood pressure can lead to narrowing of the small renal arteries, which worsens the kidney's filtering function. These alterations decrease blood delivery to the kidneys, and with time those changes can progress to **chronic kidney disease (CKD).**

Eyes

Persistently elevated blood pressure results in reduction of the lumen of retinal arteries from atherosclerotic vessel narrowing and decreased flow. With time, this leads to such vessel changes as intimal thickening (arteriovenous nicking), microaneurysms, and small retinal hemorrhages, which eventually cause vision loss from **hypertensive retinopathy** (Figure 4).



QUIZ

Figure 4. Hypertensive Retinopathy. Pathologic features include retinal hemorrhages, arteriovenous nicking, and cotton-wool spots (which represents areas of ischemic injury to the retina)

How Is Primary Hypertension Diagnosed? (LO #4)

Most cases of primary hypertension are diagnosed incidentally on routine physical examination of asymptomatic patients. Diagnostic workup includes confirming the elevated blood pressure, ruling out secondary causes, and looking for evidence of end-organ damage.

History and Physical Examination

A diagnosis of hypertension can be established with an average of at least two elevated blood pressure readings taken at least 2 weeks apart. An average blood pressure level more than 130/90 mm Hg is diagnostic for hypertension.

CLINICAL CORRELATION

Ambulatory (home-determined) blood pressure readings can be used in patients suspected of "white coat" hypertension, in which blood pressure only elevates during the clinical encounter. Note that children younger than age 15 years have lower thresholds for high blood pressure.

The purpose of the rest of the physical exam is two-fold: 1) to establish whether the hypertension is causing end-organ damage (acute or chronic), then 2) to determine whether the hypertension may be the result of an underlying disease. The physical exam is normal in most patients, unless their hypertension has been long-standing and/or severe, leading to signs of heart failure (e.g. S_3, S_4, jugular venous distension, edema), strokes (e.g. neurologic findings), or eye disease (e.g. flame hemorrhages or other vascular changes on fundoscopy).

CLINICAL CORRELATION

Flame hemorrhages hat occurs in the superficial layer of the retina, characterized by their feather or flame shape. Both flame hemorrhages and retinal hemorrhages are signs of hypertensive retinopathy.

The main physical exam approach to evaluate for secondary hypertension is to listen for abdominal bruits (sounds heard in renal artery stenosis) or look for signs of cortisol excess (Cushing disease) or thyroid disease.

Laboratory tests

Table 2

Test	Reason for test ordered
Complete Blood Count	Anemia
Complete metabolic profile	Renal Function, metabolic syndrome, hyperparathyroidism
Urine analysis	Chronic Kidney Disease (CKD), Nephrotic and nephritic syndrome
Chest X-ray	End organ effects of hypertension, aorta dilation, pulmonary edema
12-lead ECG	Hypertrophy (Left ventricular hypertrophy)

CLINICAL REASONING

Some other labs to c m would be polysomnogram (for d (for Renovascular hypertension), Urine drug screen, thyroid-stimulating hormone (hyper and hypothyroidism), and 24-hour urine for metanephrines (metabolites of catecholamines, a test used to evaluate for pheochromocytoma, an adrenal tumor that secretes excess catecholamines and causes episodic hypertension).

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to CJ, what are some potential underlying pathophysiologic mechanisms of his elevated blood pressure? How will you explain the importance of blood pressure control to him?

You know that several common mechanisms underly primary hypertension, but you focus your discussion with CJ on the clinical manifestations. Even though CJ is asymptomatic, his elevated blood pressure puts him at risk for serious complications in the future. "I think it's great that you watch your weight, exercise, and limit your salt intake, but sometimes that's not enough and medication is necessary," you begin. "If we don't lower your blood pressure, you may develop heart damage, blood vessel narrowing, stroke, chronic kidney disease, and vision loss." CJ replies, "If I hear you right, it seems like if I take medication now, I'll stay as healthy as possible in the future?" "That sums it up pretty well," you agree. "That sounds like a good tradeoff," he says. You tell CJ you'll work together to find the best medication for him.

Summary

- Hypertension is d which increases the risk of heart failure, stroke, and other serious health issues.
- Hypertension is diagnosed with at least two elevated blood pressure readings taken at least 2 weeks apart, with stage 1 defined as systolic 130-139 mm Hg and/or diastolic 80-89 mm Hg.
- Hypertension can be primary (idiopathic) or secondary (caused by an underlying disease), with primary hypertension accounting for 95% of cases.
- Risk factors for hypertension include age, genetics, obesity, lack of physical activity, and other chronic conditions such as diabetes and chronic kidney disease.
- Patients with hypertension are usually asymptomatic, but hypertensive emergencies can present with severe symptoms like chest pain and altered mental status.
- Chronic uncontrolled hypertension can lead to complications such as left ventricular hypertrophy, coronary artery disease, chronic kidney disease, and hypertensive retinopathy.

Review Questions

1. A 67-year-old male presents to the emergency department with blurry vision and pressure-like chest pain for 2 hours. His blood pressure is 200/115 mm Hg, and an ECG shows evidence of myocardial infarction. Which of the following best describes his elevated blood pressure?

   A. ● Hypertensive emergency
   B. ○ Seconda
   C. ○ Stage 1 hypertension
   D. ○ Stage 2 hypertension

Hide Explanation

The correct answer is hypertensive emergency (A), which is a systolic blood pressure ≥180 mm Hg and/or diastolic blood pressure ≥120 mm Hg plus signs of acute organ injury, which this patient presents with (ie, blurred vision and chest pain). Secondary hypertension (B) is due to an underlying disease, which has not been established here. Stage 1 hypertension (C) is a blood pressure of 130-139/80-89 mm Hg. Stage 2 hypertension (D) is a systolic blood pressure ≥140 mm Hg and/or diastolic blood pressure ≥90 mm Hg.

2. Which of the following is the LEAST likely complication of hypertension?

   A. ○ Atherosclerosis
   B. ○ Chronic kidney disease
   C. ● Hypothyroidism
   D. ○ Left ventricular hypertrophy
   E. ○ Stroke

Hide Explanation

The correct answer is hypothyroidism (C). Hypothyroidism is not a complication of hypertension, whereas all the other answers are complications of uncontrolled hypertension. Atherosclerosis (A) is worsened by hypertension, as hypertension causes remodeling and hypertrophy of smooth muscle cells in arteries. Chronic kidney disease (B) can result from longstanding poorly controlled hypertension due to damage to small renal arteries and the glomerulus. In left ventricular hypertrophy (D), chronic high afterload can cause the left ventricular wall to thicken over time. Stroke (E) is due to lipohyalinosis of small central nervous system arteries as well as bleeding from microaneurysms.

3. What is the most common presenting complaint for patients with stage I hypertension?

   A. ○ Chest pain
   B. ○ Blurry vision
   C. ○ Altered mental status
   D. ○ Shortness of breath
   E. ● Patients are asymptomatic (without complaint)

Hide Explanation

The correct answer is the patient would be asymptomatic (E). Early hypertension, prior to the development of end-organ damage, is most commonly asymptomatic, which can lead to a delay in diagnosis and treatment. Chest pain (A), blurry vision (B), altered mental status (C), and shortness of breath (D) are all common features of hypertensive emergency, characterized by higher pressure values (> 180/120 mm Hg) and the development of end-organ damage.

4. Which finding suggests chronic hypertension-related end-organ damage resulting in heart failure?

   A. ● S3 heart sound
   B. ○ Flame hemorrhages
   C. ○ Edema
   D. ○ Neurologic findings

Hide Explanation

The correct answer is S3 heart sound (A). An S3 heart sound can indicate heart failure, a result of chronic hypertension. Flame hemorrhages (B) are signs of eye disease, not specifically heart failure. Edema (D) is a symptom of heart failure but not exclusive to hypertension. Neurologic findings (E) suggest stroke, which can be a complication of hypertension but not specifically end-organ damage.

How would you rate this Brick?

References

- [Image]. Accessed November 11, 2022. https://static-content.springer.com/esm/art%3A10.1186%2F1471-2407-12-231/MediaObjects/12885_2012_3193_MOESM1_ESM.jpeg
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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the different types of hypertension.~~
02 ☑ ~~Describe the clinical presentation of hypertension.~~

References

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Review Learning Objectives
Check off the objectives when you feel comfortable with the concept
01 ☑ ~~Describe the different types of hypertension.~~
02 ☑ ~~Describe the clinical presentation of hypertension.~~

03 ☑ ~~Describe the common complications of hypertension.~~
04 ☑ ~~Interpret clinical findings to properly diagnose primary hypertension.~~

Objective Confidence: 4/4